

Here are 50 multiple-choice questions (MCQs) on food chemistryµbiology

1. **Which of the following is a monosaccharide?**

- a) Sucrose
- b) Maltose
- c) Glucose
- d) Lactose

Answer: c) Glucose

2. **What is the general formula of monosaccharides?**

- a) $C_nH_{2n+2}O_n$
- b) $C_nH_{2n}O_{n-2}$
- c) $C_nH_{2n}O_n$
- d) C_2H_2O

Answer: c) $C_nH_{2n}O_n$

3. **Which of the following is another name for glucose?**

- a) Dextrose
- b) Levulose
- c) Galactose
- d) Disaccharide

Answer: a) Dextrose

4. **Which carbohydrate is commonly referred to as 'table sugar'?

- a) Glucose

b) Fructose

c) Sucrose

d) Lactose

****Answer:**** c) Sucrose

5. ****Which of the following is a disaccharide?****

a) Glucose

b) Fructose

c) Sucrose

d) Galactose

****Answer:**** c) Sucrose

6. ****Which amino acid contains sulfur?****

a) Glycine

b) Methionine

c) Alanine

d) Lysine

****Answer:**** b) Methionine

7. ****What is the primary structure of a protein?****

a) Sequence of amino acids

b) Alpha helix and beta sheet

c) 3D folded structure

d) Quaternary structure

****Answer:**** a) Sequence of amino acids

8. **Which of the following is a nitrogen-containing compound found in lecithin?**

- a) Guanine
- b) Ethanolamine
- c) Choline
- d) Ammonium acetate

Answer: c) Choline

9. **The nitrogen-containing compound in cephalin is:**

- a) Guanine
- b) Ethanolamine
- c) Choline
- d) Ammonium acetate

Answer: b) Ethanolamine

10. **Which protein is found in egg whites?**

- a) Ovomucoid
- b) Casein
- c) Albumin
- d) Gluten

Answer: c) Albumin

11. **Which of the following fatty acids is considered essential?

- a) Oleic acid

- b) Stearic acid
- c) Linoleic acid
- d) Palmitic acid

****Answer:**** c) Linoleic acid

12. ****What is the chemical structure of a triglyceride?****

- a) Glycerol and three fatty acids
- b) Glycerol and two fatty acids
- c) Glycerol and one fatty acid
- d) Glycerol and four fatty acids

****Answer:**** a) Glycerol and three fatty acids

13. ****Which process converts liquid oils into solid fats?****

- a) Emulsification
- b) Hydrogenation
- c) Saponification
- d) Hydrolysis

****Answer:**** b) Hydrogenation

14. ****Triglycerides that are solid at room temperature are often referred to as:****

- a) Fats
- b) Oils
- c) Steroids
- d) Waxes

****Answer:**** a) Fats

15. **Triglycerides that are liquid at room temperature are often referred to as:**

- a) Fats
- b) Oils
- c) Steroids
- d) Waxes

Answer: b) Oils

16. **Which vitamin is also known as tocopherol?**

- a) Vitamin A
- b) Vitamin D
- c) Vitamin E
- d) Vitamin K

Answer: c) Vitamin E

17. **Which vitamin is primarily responsible for blood clotting?**

- a) Vitamin C
- b) Vitamin D
- c) Vitamin E
- d) Vitamin K

Answer: d) Vitamin K

18. **Which vitamin is found in citrus fruits and is important for collagen synthesis?**

- a) Vitamin A
- b) Vitamin B6
- c) Vitamin B12
- d) Vitamin C

****Answer:**** d) Vitamin C

19. ****Which mineral is crucial for the function of hemoglobin?****

- a) Iron
- b) Calcium
- c) Magnesium
- d) Potassium

****Answer:**** a) Iron

20. ****Which vitamin is also known as niacin?****

- a) Vitamin A
- b) Vitamin B3
- c) Vitamin B12
- d) Vitamin D

****Answer:**** b) Vitamin B3

21. ****Which chemical is used in toothpaste to prevent tooth decay?****

- a) Sodium fluoride
- b) Calcium carbonate
- c) Hydrogen peroxide
- d) Potassium chloride

****Answer:**** a) Sodium fluoride

22. ****Which of the following is used as a preservative in canned foods?****

- a) Ascorbic acid
- b) Nitrogen
- c) Sulfur dioxide
- d) Sodium chloride

****Answer:**** c) Sulfur dioxide

23. ****Which of the following is used to soften hard water?****

- a) Sodium bicarbonate
- b) Calcium chloride
- c) Sodium carbonate
- d) Sodium phosphate

****Answer:**** c) Sodium carbonate

24. ****Which substance is responsible for the fizz in carbonated drinks?****

- a) Carbon dioxide
- b) Nitrogen
- c) Oxygen
- d) Hydrogen

****Answer:**** a) Carbon dioxide

25. ****Which of the following is a synthetic dye used in food products?****

- a) Indigo

b) Tartrazine

c) Eosin

d) Ponceau

****Answer:**** b) Tartrazine

26. ****Which process involves the conversion of sugar into alcohol?****

a) Fermentation

b) Oxidation

c) Hydrolysis

d) Condensation

****Answer:**** a) Fermentation

27. ****Which process gives browned color and unique flavor to cooked food?****

a) Glycolysis

b) Maillard reaction

c) Saponification

d) Denaturation

****Answer:**** b) Maillard reaction

28. ****Which gas is commonly used in packaging of potato chips to prevent rancidity?****

a) Oxygen

b) Nitrogen

c) Carbon dioxide

d) Hydrogen

****Answer:**** b) Nitrogen

29. ****Pasteurization is a method used to:****

- a) Increase fat content
- b) Kill pathogens in milk
- c) Enhance flavor
- d) Reduce sugar

****Answer:**** b) Kill pathogens in milk

30. ****Which chemical reaction is involved in making soap from fats?****

- a) Polymerization
- b) Hydrogenation
- c) Saponification
- d) Fermentation

****Answer:**** c) Saponification

31. ****Which enzyme breaks down starch into maltose?****

- a) Lactase
- b) Amylase
- c) Pepsin
- d) Lipase

****Answer:**** b) Amylase

32. ****Which enzyme is used in cheese production?****

- a) Invertase

- b) Rennin
- c) Catalase
- d) Amylase

****Answer:**** b) Rennin

33. ****Which of the following enzymes breaks down proteins?****

- a) Lipase
- b) Pepsin
- c) Amylase
- d) Maltase

****Answer:**** b) Pepsin

34. ****The enzyme that catalyzes the breakdown of hydrogen peroxide is:****

- a) Peroxidase
- b) Oxidase
- c) Catalase
- d) Amylase

****Answer:**** c) Catalase

35. ****Lipase acts on which class of biomolecules?****

- a) Carbohydrates
- b) Proteins
- c) Lipids
- d) Nucleic acids

****Answer:**** c) Lipids

36. **Aflatoxins are produced by:**

- a) Bacteria
- b) Viruses
- c) Fungi
- d) Protozoa

Answer: c) Fungi

37. **Which of the following is a heavy metal contaminant in food?**

- a) Zinc
- b) Iron
- c) Mercury
- d) Calcium

Answer: c) Mercury

38. **Which contaminant is commonly found in stored grains?**

- a) Lead
- b) Salmonella
- c) Aflatoxin
- d) Botulinum

Answer: c) Aflatoxin

39. **Botulism is caused by a toxin from:**

- a) Clostridium botulinum
- b) Salmonella typhi

c) *E. coli*

d) *Bacillus subtilis*

****Answer:**** a) *Clostridium botulinum*

40. ****Which chemical is used to detect adulteration of turmeric with metanil yellow?****

a) Sulfuric acid

b) Hydrochloric acid

c) Nitric acid

d) Acetic acid

****Answer:**** c) Nitric acid

41. ****Caffeine is most abundantly found in:****

a) Soft drinks

b) Coffee

c) Milk

d) Juice

****Answer:**** b) Coffee

42. ****Which compound gives pungency to mustard and radish?****

a) Allyl isothiocyanate

b) Capsaicin

c) Eugenol

d) Citral

****Answer:**** a) Allyl isothiocyanate

43. **Which acid is naturally present in citrus fruits?**

- a) Acetic acid
- b) Tartaric acid
- c) Lactic acid
- d) Citric acid

Answer: d) Citric acid

44. **The main acid found in vinegar is:**

- a) Citric acid
- b) Tartaric acid
- c) Acetic acid
- d) Lactic acid

Answer: c) Acetic acid

45. **Capsaicin is found in:**

- a) Black pepper
- b) Red chili
- c) Ginger
- d) Turmeric

Answer: b) Red chili

46. **Which of the following is considered a probiotic?**

- a) Lactic acid
- b) Lactobacillus

c) Casein

d) Ascorbic acid

****Answer:**** b) Lactobacillus

47. ****Which of the following is an example of a functional food?****

a) White bread

b) Sugar

c) Yogurt with live cultures

d) Soda

****Answer:**** c) Yogurt with live cultures

48. ****Omega-3 fatty acids are beneficial for:****

a) Skin only

b) Brain and heart health

c) Bone strength

d) Hair color

****Answer:**** b) Brain and heart health

49. ****Which of the following is a plant pigment with antioxidant properties?****

a) Casein

b) Anthocyanin

c) Glucose

d) Maltose

****Answer:**** b) Anthocyanin

50. **Which phytochemical is abundant in tomatoes and linked to cancer prevention?**

- a) Lycopene
- b) Lutein
- c) Zeaxanthin
- d) Resveratrol

Answer: a) Lycopene

1. Which of the following microorganisms is primarily responsible for milk fermentation?

- A. Escherichia coli
- B. Lactobacillus spp.
- C. Salmonella spp.
- D. Clostridium botulinum

Answer: B

2. Which bacterium is most commonly associated with foodborne botulism?

- A. Bacillus cereus
- B. Clostridium perfringens
- C. Clostridium botulinum
- D. Listeria monocytogenes

Answer: C

****3. Which of the following is a spore-forming bacterium?****

- A. *Listeria monocytogenes*
- B. *Salmonella enterica*
- C. *Clostridium botulinum*
- D. *Escherichia coli*

****Answer:** C**

****4. Pasteurization is used to:****

- A. Sterilize food
- B. Improve taste
- C. Kill pathogens and reduce spoilage organisms
- D. Add nutrients

****Answer:** C**

****5. The optimal temperature for the growth of most foodborne pathogens is:****

- A. 0–4°C
- B. 25–35°C
- C. 60–75°C
- D. -20°C

****Answer:** B**

****6. Which microorganism is commonly used in bread making?****

- A. *Lactobacillus*
- B. *Rhizopus*
- C. *Saccharomyces cerevisiae*

D. Penicillium

****Answer:** C**

****7. Which of the following causes “blue mold” on bread?****

A. Aspergillus

B. Penicillium

C. Cladosporium

D. Fusarium

****Answer:** B**

****8. A flat-sour spoilage in canned food is caused by:****

A. Lactobacillus

B. Clostridium botulinum

C. Bacillus stearothermophilus

D. Saccharomyces

****Answer:** C**

****9. The presence of coliforms in food indicates:****

A. Food is sterile

B. Presence of fungi

C. Possible fecal contamination

D. Spoilage due to yeast

****Answer:** C**

****10. Which method is best for sterilizing heat-sensitive food?****

- A. Boiling
- B. Autoclaving
- C. Filtration
- D. Radiation

****Answer:** C**

****11. Which bacteria are most commonly linked with raw poultry?****

- A. Listeria
- B. Salmonella
- C. Staphylococcus
- D. Vibrio

****Answer:** B**

****12. What is the infectious agent in prion diseases found in food (e.g., BSE)?****

- A. Bacteria
- B. Virus
- C. Misfolded proteins
- D. Fungi

****Answer:** C**

****13. The HACCP system is primarily designed to:****

- A. Improve taste
- B. Manage food cost
- C. Prevent food safety hazards

D. Enhance shelf life

****Answer:** C**

****14. Which microorganism is associated with listeriosis?****

A. Clostridium

B. Salmonella

C. *Listeria monocytogenes*

D. *Vibrio cholerae*

****Answer:** C**

****15. A toxin-producing bacteria that can grow in improperly stored rice is:****

A. *E. coli*

B. *Bacillus cereus*

C. *Campylobacter jejuni*

D. *Staphylococcus aureus*

****Answer:** B**

****16. The term “psychrotrophic” refers to organisms that:****

A. Thrive at very high temperatures

B. Grow best in acidic food

C. Grow at refrigeration temperatures

D. Require high salt concentrations

****Answer:** C**

****17. Which of the following is a chemical preservative?****

- A. Vinegar
- B. Sodium benzoate
- C. Sugar
- D. Salt

****Answer:** B**

****18. Fermentation helps in:****

- A. Increasing pathogenic load
- B. Spoiling the food
- C. Enhancing flavor and preserving food
- D. Reducing nutrients

****Answer:** C**

****19. Which gas is often used in modified atmosphere packaging (MAP) to inhibit microbial growth?****

- A. Oxygen
- B. Nitrogen
- C. Carbon dioxide
- D. Methane

****Answer:** C**

****20. Which of these microbes is tolerant to high salt concentrations?****

- A. E. coli
- B. Clostridium

C. Halophilic bacteria

D. Bacillus

****Answer:** C**

****21. Yogurt is made by fermenting milk with:****

A. Yeast

B. Acetobacter

C. Lactobacillus and Streptococcus

D. Penicillium

****Answer:** C**

****22. Which microorganism is used in beer production?****

A. Candida albicans

B. Rhizopus

C. Saccharomyces cerevisiae

D. E. coli

****Answer:** C**

****23. The process of converting sugars to alcohol anaerobically is known as:****

A. Pasteurization

B. Fermentation

C. Respiration

D. Oxidation

****Answer:** B**

****24. Which acid is produced during sauerkraut fermentation?****

- A. Acetic acid
- B. Citric acid
- C. Lactic acid
- D. Malic acid

****Answer:** C**

****25. Nisin is an example of a:****

- A. Enzyme
- B. Antibiotic
- C. Bacteriocin
- D. Preservative

****Answer:** C**

****26. Thermophilic bacteria thrive best at:****

- A. 0–10°C
- B. 20–30°C
- C. 50–60°C
- D. 100°C

****Answer:** C**

****27. Which of the following is used to detect foodborne pathogens in the lab?****

- A. Gram staining
- B. ELISA
- C. Pasteurization

D. Filtration

****Answer:** B**

****28. Aflatoxins are produced by:****

A. Rhizopus

B. Aspergillus flavus

C. Penicillium

D. Clostridium

****Answer:** B**

****29. Food irradiation is primarily used to:****

A. Cook food

B. Improve taste

C. Kill pathogens and spoilage organisms

D. Increase shelf temperature

****Answer:** C**

****30. Campylobacter is commonly found in:****

A. Eggs

B. Milk

C. Raw chicken

D. Canned foods

****Answer:** C**

****31. Which bacterium is known for producing enterotoxins in improperly stored cream or custard?****

- A. *Bacillus cereus*
- B. *Staphylococcus aureus*
- C. *Listeria monocytogenes*
- D. *Campylobacter jejuni*

****Answer:** B**

****32. Which fungus is commonly involved in the spoilage of fruits?****

- A. *Rhizopus stolonifer*
- B. *Candida albicans*
- C. *Clostridium perfringens*
- D. *Staphylococcus aureus*

****Answer:** A**

****33. Which of the following is NOT a method of food preservation?****

- A. Irradiation
- B. Drying
- C. Fermentation
- D. Contamination

****Answer:** D**

****34. Which microorganism is used in the production of vinegar?****

- A. *Acetobacter aceti*
- B. *Saccharomyces cerevisiae*

C. *Lactobacillus bulgaricus*

D. *Penicillium chrysogenum*

****Answer:** A**

****35. The major risk of undercooked ground beef is infection by:****

A. *Listeria*

B. *E. coli* O157\ :H7

C. *Salmonella typhi*

D. *Bacillus subtilis*

****Answer:** B**

****36. Which of the following organisms can grow at refrigeration temperatures and cause disease?****

A. *Clostridium botulinum*

B. *Listeria monocytogenes*

C. *E. coli*

D. *Campylobacter*

****Answer:** B**

****37. Which bacterium is commonly associated with seafood and causes cholera-like symptoms?****

A. *Salmonella*

B. *Vibrio vulnificus*

C. *Bacillus cereus*

D. *Shigella dysenteriae*

****Answer:** B**

****38. Which of the following methods is best for killing endospores?****

- A. Boiling
- B. Freezing
- C. Autoclaving
- D. Pasteurization

****Answer:** C**

****39. The pH range at which most bacteria cannot grow is:****

- A. 4.5–7.5
- B. Below 4.5
- C. 6.0–8.0
- D. Above 9.0

****Answer:** B**

****40. The term "food intoxication" refers to:****

- A. Infection caused by viruses in food
- B. Toxin production in food before consumption
- C. Poisoning due to allergic reaction
- D. Infection from parasites

****41. Which of the following is NOT a foodborne virus?****

- A. Norovirus
- B. Hepatitis A

C. Rotavirus

D. Vibrio cholerae

****Answer:**** D

****42. Which of the following techniques is used to enumerate total viable bacteria in a sample?****

A. Gram staining

B. Pour plate method

C. ELISA

D. PCR

****Answer:**** B

****43. The "danger zone" in food safety typically refers to temperatures between.****

A. 0–4°C

B. 5–60°C

C. 70–90°C

D. 100°C

****Answer:**** B

****44. Which microbial group is often responsible for spoilage of high-acid foods like fruit juices?****

A. Bacteria

B. Yeasts and molds

C. Protozoa

D. Algae

****Answer:** B**

****45. Indicator organisms in microbiological testing are used to:****

- A. Detect viruses
- B. Estimate contamination and sanitary quality
- C. Improve taste
- D. Remove toxins

****Answer:** B**

****46. Which of the following bacteria is anaerobic and forms spores?****

- A. Listeria
- B. Salmonella
- C. Clostridium perfringens
- D. Shigella

****Answer:** C**

****47. What is the function of nitrites in meat preservation?****

- A. Enhance flavor
- B. Inhibit Clostridium botulinum
- C. Improve color only
- D. Prevent mold growth

****Answer:** B**

****48. The role of probiotics in food microbiology is to:****

- A. Cause fermentation

- B. Spoil food
- C. Promote gut health by providing beneficial bacteria
- D. Preserve food

****Answer:** C**

****49. A psychrophile is a microorganism that thrives at:****

- A. High pressure
- B. High pH
- C. Cold temperatures
- D. Low oxygen levels

****Answer:** C**

****50. Which of the following organisms is a common cause of traveler's diarrhea?****

- A. Campylobacter
- B. Listeria
- C. E. coli (ETEC)
- D. Staphylococcus aureus

****Answer:** C**